EXPERIMENT – 12

Extraction of EXIF Data from Image Files using ExifReader

Aim

To extract the Exchangeable image file format (EXIF) data from an image file using the ExifReader software.

Introduction

When a computer, phone or mobile device is seized for evidence, the system usually contains graphic images that may be used as evidence. Most digital devices 'stamp' information on these images that can tell us about the who, what, when and where of the picture. This information is known as EXIF data and is often useful to the forensic investigator.

EXIF is a standard set by the digital camera industry to identify formats for digital images and sound files. It includes camera settings, time, date, shutter speed, exposure, whether a flash was used, compression, the name of the camera and other information. Several applications can extract this data; for this experiment a simple Windows-based tool called ExifReader is used.

Procedure

- Download ExifReader and run the file ExifRead.exe. It opens a clean and simple GUI wizard.
- Click on the 'Open' button and browse to the picture stored on the system or media. JPEG and JPG files normally contain the maximum information.
- Once the selected picture opens, it is loaded as a thumbnail on the left and the EXIF data is displayed on the right.
- Study the extracted information such as the Make and Model of the camera, DateTime, DateTimeOriginal, Exposure Time, FNumber, Exposure Program, ISO Speed Ratings, Compressed Bits, Focal Length, Flash and the GPS location.
- Note the details that are relevant to the investigation, as EXIF data can reveal when and where an image was captured and with which device.

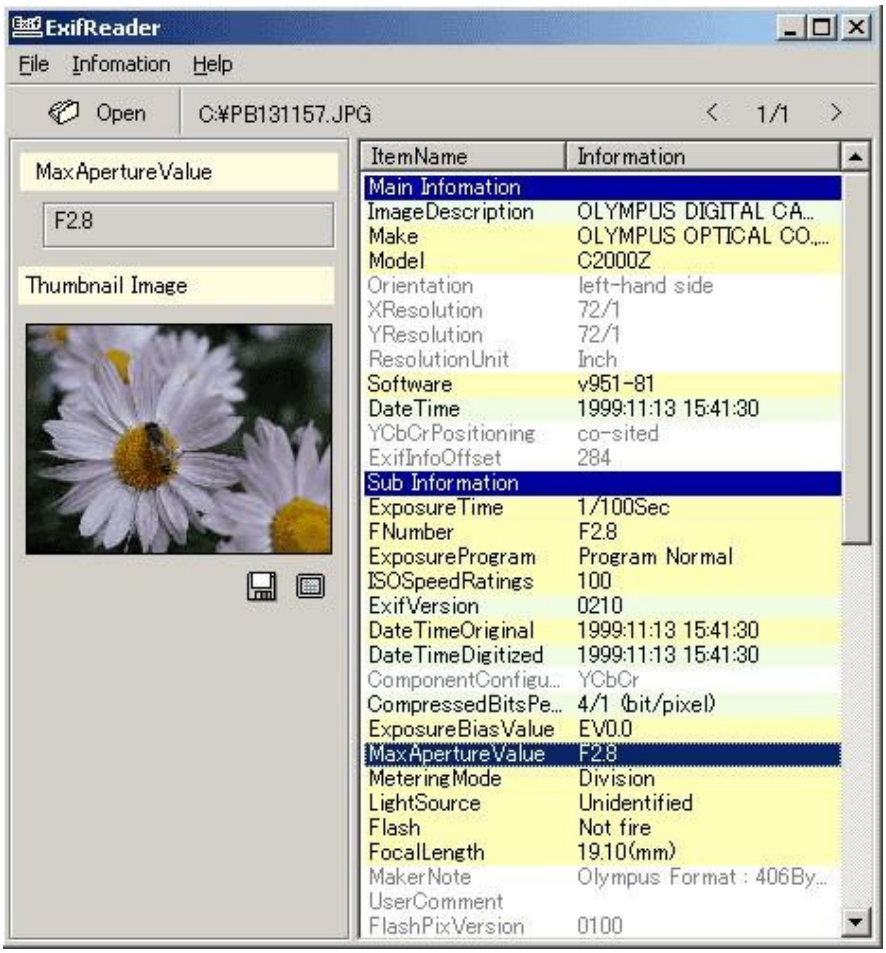


Fig. 12.1 ExifReader – EXIF metadata extracted from a JPEG image
Timestamp: 13-10-2026, 16:05:39

Result

Thus, the EXIF data was extracted successfully from the image file using the ExifReader software. The camera make and model, the date and time the picture was taken and the photographic settings were obtained, which can help the forensic investigator to link the image to its source.